VBF-T7R3-DS-01 Case t7_r3 | 2026-08-26

Prepared: ______________ Reviewed: ______________ Approved: ______________

Virtual design package - simulation caliber (no physical build)

VBF Design Specification - HEV Battery Cell (virtual design)

Document number: VBF-T7R3-DS-01 · **Case**: t7_r3 · **Date**: 2026-08-26
Base parameter set: Chen2020 (PyBaMM) · **Design candidate**: v19_final - `candidates/r7_v19_final_params.json`
Status: virtual design (simulation caliber; no physical build). Prepared / Reviewed / Approved: ________ (blank for signature)

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1. Basic specification

Field	Value	Source
Electrochemical system	NMC811 (positive) / graphite (negative), LiPF■-class carbonate electrolyte	Chen2020 base set
Nominal capacity	6.28 Ah (set), simulated 1C capacity 6.2569 Ah (SPMe) / 6.28 Ah (DFN)	`r7_v19_final_1c.json:capacity_ah`, `r6_v19_1c_dfn.json`
Voltage window	2.5 V – 4.2 V	Chen2020 `Upper/Lower voltage cut-off [V]`
Electrode dimensions	height 0.065 m × width 1.58 m (unwound strip); stack thickness 225.6 μm (pos 75.6 + sep 12 + neg 110 + Al 16 + Cu 12 μm)	Chen2020 geometry + `r7_v19_final_energy.json:thickness_m`
Shell / case thickness	**Not provided** (no parameter in set)	Chen2020
Electrolyte formulation	carrier: Chen2020 electrolyte functions replaced by fractional values: **cation transference number 0.6**, **diffusivity 1.0×10-9 m^2/s**, conductivity 1.5 S/m	`r7_v19_final_params.json` (formulation **estimate** - see Note 1)
SEI additive candidate	FEC (film-forming additive) - proposed in plan, **not simulated** (Stage 2 skipped, `meta.start_stage=3`)	`log.jsonl` entry 0 meta

Note 1: transference number 0.6 / D■ 1×10-9 m^2/s are **formulation-level estimates**, not a specific salt/solvent recipe. Literature data (Valøen & Reimers 2005, J. Electrochem. Soc.) show t+ ~ 0.26–0.4 for conventional LiPF■/EC systems; concentrated or salt-tuned electrolytes and fluorinated ether blends reach t+ 0.5–0.7. A real formulation must be down-selected and measured; the values here are the cell-design target properties.

2. Electrode and separator

Layer	Thickness (μm)	Porosity	Active volume fraction	Particle radius (μm)	Inventive value
Positive (NMC811)	75.6	0.335	0.665	**2.0** (base 5.22)	`r7_v19_final_params.json`
Negative (graphite)	**110** (base 85.2)	0.25	0.75	**2.0** (base 5.86)	`r7_v19_final_params.json`
Separator	12	0.47	-	-	Chen2020

Current collector	Thickness (μm)	Density (kg/m^3)	Source
Positive (Al)	16	2700	Chen2020
Negative (Cu)	12	8960	Chen2020

N/P ratio = (Q_neg × eps_s,neg × L_neg) / (Q_pos × eps_s,pos × L_pos) = (33133 × 0.75 × 110×10-■) / (63104 × 0.665 × 75.6×10-■) = **0.862** (mechanical from Chen2020 max concentrations + layer geometry). Note: below the 1.05–1.15 industry practice for graphite anodes; the virtual design passes 4C plating at this balance - see §6.

3. Process design parameters

Parameter	Value	Formula / source
Positive areal density	164.0 g/m^2	thickness×(1−eps)×density = 75.6 μm×0.665×3262 kg/m^3 (energy contract)
Negative areal density	136.7 g/m^2	110 μm×0.75×1657 kg/m^3 (energy contract)
Positive compaction density	2.17 g/cm^3	3262×(1−0.335) kg/m^3 ÷ 1000
Negative compaction density	1.24 g/cm^3	1657×(1−0.25) kg/m^3 ÷ 1000
Electrolyte fill amount	7.21 g/cell (~6.00 mL)	pore volume 6.004×10-■ m^3 (SUM Lt×eps×A) × 1200 kg/m^3 (literature density 1.2 g/cm^3, annotated default), fill factor 1.0
Formation recommendation	0.1C CC to 4.2 V, 25 °C, 2 cycles	**design-recommended value; actual production parameter requires line tuning**

4. Mass breakdown (contract caliber: electrolyte excluded)

Layer	kg/m^2	g/cell (×0.1027 m^2)	Source
Positive electrode	0.1640	16.84	`r7_v19_final_energy.json:layer_kg_m2`

Negative electrode	0.1367	14.04	same
Positive CC (Al)	0.0432	4.44	same
Negative CC (Cu)	0.1075	11.04	same
Separator	0.00252	0.259	same
Sum (contract)	-	**46.62 g**	`mass_kg = 0.0466198`
Electrolyte (additive)	-	+7.21 g	literature density, see §3
Total incl. electrolyte	-	**53.83 g**	derived

5. Performance verification vs criteria (entry 0)

Criterion	Threshold	Result	Verdict	Evidence
Energy density	>= 327.18 Wh/kg	**482.4 Wh/kg** (contract; electrolyte excluded) / 417.9 Wh/kg incl. electrolyte	[PASS] PASS	`r7_v19_final_energy.json:energy_density_wh_kg`
SEI after 100 cyc @45 °C	<= 550 nm	**512.9 nm**	[PASS] PASS	`r7_v19_final_aging45c.json:sei_thickness_nm_end`
4C charge plating	plated = false	anode_potential_v min = **+0.0171 V** (0 negative instants)	[PASS] PASS	`r7_v19_final_4c45c.json:anode_potential_v`
Nail 10 W thermal runaway	triggered = false	**false**, T_max 318.2 K @ hA = 0.5 W/K	[PASS] PASS	`r7_v19_final_nail_hA05.json`

Additional measured values: 1C discharge capacity 6.2569 Ah (SPMe); rated energy 22.491 Wh; midpoint voltage 4.047 V; DCR 0.120 mohm; volumetric energy density 970.7 Wh/L; 4C CC charge accepts 0.778 Ah (12.4 % SOC) before 4.2 V cutoff, T_max during 4C 363.2 K.

6. Design notes (what changed and why - from evaluate log)

Parameter	Base -> Final	Reason (log)
Negative electrode thickness	85.2 -> 110 µm	capacity/ED balance toward 6.28 Ah nominal
Cation transference number	0.2594 -> 0.6	electrolytic concentration polarization was the 4C plating root cause (R4 V9 flipped plated true->false)
Electrolyte diffusivity	Nyman2008 fn -> 1.0×10-9 m^2/s	same root cause; combination with t+ gives full CC window margin
Electrolyte conductivity	fn -> 1.5 S/m	minor contribution (R2/R4: sigma alone inert for plating)
Positive particle radius	5.22 -> 2.0 µm	charge-depth limiter after electrolyte fix = positive solid diffusion (Chen2020 D_s,pos = 4×10-^{1}■ m^2/s)
Negative particle radius	5.86 -> 2.0 µm	anode surface-area margin for plating robustness
Nominal cell capacity	5.0 -> 6.28 Ah	honesty correction: rating set to simulated 1C capacity -> true 4C = 25.12 A

Thermal-management design requirement (from nail sweep): active cooling with **hA >= 0.3 W/K** (measured minimum passing), design point **hA = 0.5 W/K** (T_max 318.2 K). At hA = 0.2 W/K the cell triggers (T = 405 K at 2097 s); at 0.05 W/K near-adiabatic it triggers at 346 s. hA = 0.5 W/K corresponds to ~2.4 W/m^2K over the 0.205 m^2 double-sided pouch surface - forced-air level cooling, realistic for HEV pack.

N/P = 0.862 note: mechanically derived from the returned set values; the design passes 4C plating under this balance. For production intent, N/P 1.05–1.15 (thicker negative) is recommended and would further increase the +17 mV anode margin; it would trade ~5 % energy density.

Structure model (`cell_model.stl`): **not requested in this headless session** (per protocol, decided by user clarification); recommended presentation if requested later: pouch stacked, exploded view with real thickness annotations.